Electrodynamics & Optics

Part II Physics — Key Derivations

Compiled from Handouts 1–3, the lecture overheads and the examples sheet
(Prof. C. A. Haniff, Michaelmas Term 2025)

This document presents the derivations judged most important for the course — the ones most likely to be tested, that establish a central result, or that teach a transferable method. Each is preceded by a short statement of strategy. Boxed lines are the results to remember. Methods drawn from the (examinable) examples sheet are included where they add technique.

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1 The EM wave equation, impedance and transversality

Strategy: Take the curl of a curl equation and use $\nabla\times(\nabla\times\mathbf{F}) = \nabla(\nabla\cdot\mathbf{F}) - \nabla^2\mathbf{F}$ to decouple $\mathbf{E}$ and $\mathbf{B}$.

In a source-free linear medium ($\rho = 0, \mathbf{J} = 0$), take the curl of ME3 and use ME4 with $\mathbf{D} = \epsilon\epsilon_0\mathbf{E}$, $\mathbf{B} = \mu\mu_0\mathbf{H}$:

$$\begin{aligned}\nabla\times(\nabla\times\mathbf{E}) &= -\nabla\times\dot{\mathbf{B}} = -\mu\mu_0\frac{\partial}{\partial t}(\nabla\times\mathbf{H}) = -\epsilon\epsilon_0\mu\mu_0\,\ddot{\mathbf{E}},\\ \nabla(\nabla\cdot\mathbf{E}) - \nabla^2\mathbf{E} &= -\epsilon\epsilon_0\mu\mu_0\,\ddot{\mathbf{E}}.\end{aligned}$$

With $\nabla\cdot\mathbf{E} = 0$ this gives the wave equation (and similarly for $\mathbf{B}$):

$$\boxed{\nabla^2\mathbf{E} = \epsilon\epsilon_0\mu\mu_0\,\ddot{\mathbf{E}}, \qquad v = \frac{1}{\sqrt{\epsilon\epsilon_0\mu\mu_0}} = \frac{c}{n}, \quad c = \frac{1}{\sqrt{\epsilon_0\mu_0}}, \; n = \sqrt{\epsilon\mu}.}$$

For harmonic plane waves $e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}$ the operators become $\nabla \to i\mathbf{k}$, $\partial_t \to -i\omega$, so Maxwell's equations read $\mathbf{k}\cdot\mathbf{D} = 0$, $\mathbf{k}\cdot\mathbf{B} = 0$, $\mathbf{k}\times\mathbf{E} = \omega\mathbf{B}$, $\mathbf{k}\times\mathbf{H} = -\omega\mathbf{D}$. Hence $\mathbf{B}, \mathbf{D} \perp \mathbf{k}$; for isotropic media $(\mathbf{E}, \mathbf{H}, \mathbf{k})$ form a right-handed orthogonal set with $\mathbf{N} = \mathbf{E}\times\mathbf{H} \parallel \mathbf{k}$. Taking magnitudes of $\mathbf{k}\times\mathbf{E} = \omega\mathbf{B}$ gives $|\mathbf{E}|/|\mathbf{B}| = \omega/k = v$, and with $\mathbf{B} = \mu\mu_0\mathbf{H}$,

$$\boxed{\frac{|\mathbf{E}|}{|\mathbf{H}|} = \sqrt{\frac{\mu\mu_0}{\epsilon\epsilon_0}} = \sqrt{\frac{\mu}{\epsilon}}\,Z_0, \qquad Z_0 = \sqrt{\mu_0/\epsilon_0} \approx 377\,\Omega.}$$

2 Poynting's theorem

Strategy: Differentiate the energy density and substitute ME3, ME4 to expose a divergence (flux) term and a dissipation term.

With $u = \frac{1}{2}\mathbf{E}\cdot\mathbf{D} + \frac{1}{2}\mathbf{B}\cdot\mathbf{H}$ and the identity $\nabla\cdot(\mathbf{a}\times\mathbf{b}) = \mathbf{b}\cdot(\nabla\times\mathbf{a}) - \mathbf{a}\cdot(\nabla\times\mathbf{b})$,

$$\frac{du}{dt} = \mathbf{E}\cdot\dot{\mathbf{D}} + \mathbf{H}\cdot\dot{\mathbf{B}} = \mathbf{E}\cdot(\nabla\times\mathbf{H} - \mathbf{J}) - \mathbf{H}\cdot(\nabla\times\mathbf{E}) = -\nabla\cdot(\mathbf{E}\times\mathbf{H}) - \mathbf{J}\cdot\mathbf{E}.$$

Integrating over V:

$$\boxed{\frac{dU}{dt} = -\oint_S(\mathbf{E}\times\mathbf{H})\cdot d\mathbf{S} - \int_V \mathbf{J}\cdot\mathbf{E}\,dV, \qquad \mathbf{N} = \mathbf{E}\times\mathbf{H}.}$$

3 The dielectric tensor is Hermitian

Strategy: Compute du/dt two ways (direct differentiation vs. energy conservation) and equate; then time-average a product of complex harmonic quantities.

Differentiating u directly and also using energy conservation ($du/dt = -\nabla\cdot\mathbf{N} - \mathbf{J}\cdot\mathbf{E}$ and ME3, ME4) gives, after subtracting,

$$(\dot{\mathbf{E}}\cdot\mathbf{D} - \mathbf{E}\cdot\dot{\mathbf{D}}) + (\dot{\mathbf{H}}\cdot\mathbf{B} - \mathbf{H}\cdot\dot{\mathbf{B}}) = 0.$$

The dielectric and magnetic responses are independent, so each bracket vanishes: $\mathbf{E}\cdot\dot{\mathbf{D}} = \mathbf{D}\cdot\dot{\mathbf{E}}$. Taking real parts and using, for harmonic quantities, $\langle\Re(a)\Re(b)\rangle = \frac{1}{2}\Re(a^*b)$,

$$\Re\sum_{ij} E_i^*\epsilon_{ij}\dot{E}_j = \Re\sum_{ij}\dot{E}_j\epsilon_{ji}^*E_i^* \quad\Longrightarrow\quad \boxed{\epsilon_{ij} = \epsilon_{ji}^*}\ \text{(Hermitian)}.$$

A Hermitian tensor can be diagonalised; for lossless, non-active media it is real and symmetric, defining principal axes and indices.

4 Jones matrix of a polaroid at angle θ (Examples Q3)

Strategy: Rotate into the polarizer's frame, project, rotate back: $J_\theta = R(\theta)J_xR(-\theta)$.

With $R(\theta) = \begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}$ and $J_x = \mathrm{diag}(1,0)$, carrying out $R(\theta)J_xR(-\theta)$:

$$J_\theta = R(\theta)\begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix}R(-\theta) = \begin{pmatrix}\cos^2\theta & \sin\theta\cos\theta\\ \sin\theta\cos\theta & \sin^2\theta\end{pmatrix}.$$

Acting on L_x and computing $|J_\theta L_x|^2$ recovers **Malus's law** $I(\theta) = I_0\cos^2\theta$. For sequential elements A, B, C the overall matrix is $J = J_CJ_BJ_A$; crossed polarizers give $J_\theta J_{\pi/2+\theta} = 0$.

5 Waveplate phase and the quarter-wave plate thickness (Examples Q4)

Strategy: The two principal polarizations accrue optical phases $\omega nd/c$; their difference is the retardance.

A plate of thickness d imposes $\mathrm{diag}(e^{i\omega n_f d/c}, e^{i\omega n_s d/c}) \propto \mathrm{diag}(e^{-i\Delta\phi/2}, e^{+i\Delta\phi/2})$ with $\Delta\phi = \omega(n_s - n_f)d/c$. A quarter-wave plate needs $\Delta\phi = \pi/2$, i.e. $\omega|n_e - n_o|\,d/c = \pi/2$, giving

$$d_{\min} = \frac{\pi c}{2\omega|n_e - n_o|} = \frac{\lambda}{4|n_e - n_o|}.$$

For $|n_e - n_o| \sim 0.01$–0.1 and $\lambda \sim 0.5\,\mu$m this is a few μm to tens of μm. (Acting $J_{\lambda/4}$ on $(\cos\theta, \sin\theta)$ gives $(\cos\theta, i\sin\theta)$ — elliptical, and circular for $\theta = 45°$.)

6 Optical rotation in a chiral medium

Strategy: Write linear light as equal LCP+RCP, apply the different phases $e^{i\omega n_{L,R}d/c}$, and recombine.

Using $L_x = (C_R + C_L)/\sqrt{2}$ and $\Delta\phi = \omega(n_L - n_R)d/c$,

$$\tfrac{1}{\sqrt{2}}\left[\tfrac{1}{\sqrt{2}}\begin{pmatrix}1\\ -i\end{pmatrix}e^{-i\Delta\phi/2} + \tfrac{1}{\sqrt{2}}\begin{pmatrix}1\\ i\end{pmatrix}e^{+i\Delta\phi/2}\right] = \begin{pmatrix}\cos(\Delta\phi/2)\\ -\sin(\Delta\phi/2)\end{pmatrix},$$

a plane wave rotated by $\Delta\phi/2$. The specific rotatory power is

$$\alpha = \frac{\Delta\phi}{2d} = \frac{\pi(n_L - n_R)}{\lambda} = \frac{\omega}{2c}(n_L - n_R).$$

7 Faraday rotation in a magnetised plasma

Strategy: Solve the electron equation of motion in $\mathbf{E}\,e^{-i\omega t}$ and $\mathbf{B}_0 \parallel \hat{z}$ using the circular combinations $x_0 \pm iy_0$, which diagonalise the problem into LCP/RCP responses.

For each electron, neglecting the wave's $\mathbf{B}$ and scattering, $m\ddot{\mathbf{r}} = -e(\mathbf{E}+\dot{\mathbf{r}}\times\mathbf{B}_0)$. With harmonic motion the x, y equations are

$$-\omega^2 x_0 = -\tfrac{e}{m}E_x + i\omega\omega_c y_0, \qquad -\omega^2 y_0 = -\tfrac{e}{m}E_y - i\omega\omega_c x_0, \quad \omega_c = \frac{eB_0}{m}.$$

Forming $x_0 \pm iy_0$ decouples them; the $(x_0 - iy_0)$ combination is driven by an LCP field $C_L = E(1, i)$, giving an effective susceptibility and permittivity

$$\epsilon_{L,R}(\omega) = 1 - \frac{\omega_p^2}{\omega(\omega \mp \omega_c)}, \qquad \omega_p^2 = \frac{ne^2}{\epsilon_0 m}.$$

Since $n_L \neq n_R$, a plane wave rotates by $\theta = \frac{\omega d}{2c}(n_L - n_R)$; for small B_0, $\theta \approx -\dfrac{\omega_p^2\omega_c d}{2c\omega^2}\big(1-\omega_p^2/\omega^2\big)^{-1/2}$, defining the Verdet coefficient via $\theta = VB_0 d$. Solving for x_0, y_0 generally yields the full Hermitian dielectric tensor with off-diagonal terms $\pm i\omega_c\omega_p^2/[\omega(\omega^2-\omega_c^2)]$.

8 Temporal coherence and the Wiener–Khinchine theorem

Strategy: Express the Michelson output as $\langle|A_1 + A_2|^2\rangle$, identify $\Gamma(\tau)$, then Fourier-transform the autocorrelation.

With $A_1 = f(t)$, $A_2 = f(t-\tau)$ and $\tau = 2d/c$,

$$I(\tau) = \langle|A_1 + A_2|^2\rangle = 2I_0 + 2\Re[\Gamma(\tau)], \qquad \Gamma(\tau) = \langle f(t) f^*(t-\tau)\rangle.$$

Writing $\Gamma = |\Gamma|e^{i\Delta}$, the fringe visibility is $V = (I_{\max} - I_{\min})/(I_{\max} + I_{\min}) = |\Gamma(\tau)|/I_0 = |\gamma(\tau)|$. For the autocorrelation $h(\tau) = \int f(t)f^*(t-\tau)\,dt$, a change of variable $t' = t - \tau$ gives

$$\mathrm{FT}[h(\tau)] = F(\omega)F^*(\omega) = |F(\omega)|^2 \equiv P(\omega) \quad \text{(Wiener–Khinchine)}.$$

Hence the power spectrum is the FT of the (measured) degree of coherence: $P(\omega) \sim \mathrm{FT}[\gamma(\tau)] = \mathrm{FT}[I_r(\tau) - 1]$ — the principle of Fourier-transform spectroscopy.

8.1 Worked example: Gaussian line (and coherence length)

For $P(\omega) = C\exp[-(\omega-\omega_0)^2/2\sigma^2]$, inverse-transforming (completing the square) gives

$$\Gamma(\tau) \sim e^{i\omega_0\tau}\exp\Big(-\tfrac{\sigma^2\tau^2}{2}\Big) \;\Rightarrow\; V(\tau) = e^{-\sigma^2\tau^2/2}.$$

The coherence length (path difference $2d$ at which $V = 1/e$) is $l_c = c/(\sqrt{2}\,\sigma) \sim \lambda^2/(2\delta\lambda)$, the basis of Examples Q13–Q14.

9 Spatial coherence: the van Cittert–Zernike theorem

Strategy: Sum intensities *(incoherent source) of two-slit patterns from each source point; the cross term is a Fourier transform of $I(\theta)$.*

For a source point at x and screen point y, the slit path difference gives a phase $2ks$ with $ks = \frac{1}{2}kd(x/L + y/D)$. The amplitude at P from $x \to x + dx$ is $\psi(y) \sim \sqrt{I(x)}\,2\cos ks$. For a spatially *incoherent* source we add intensities:

$$I_y(d) = \int |\psi|^2 dx = 2I_0 + 2\Re\Big[e^{-ikdy/D}\int I(x)e^{-ikdx/L}dx\Big].$$

Changing to angles ($x \approx L\theta$) and writing $u = kd$,

$$\boxed{\gamma(u) = \frac{1}{I_0}\int I(\theta)e^{-iu\theta}\,d\theta = \frac{\mathrm{FT}[I(\theta)]}{I_0} \qquad \text{(van Cittert–Zernike).}}$$

The fringe visibility is $V = \gamma(u = kd)$. **Line source** of width α: $\gamma = \mathrm{sinc}(u\alpha/2)$, first zero at $d = \lambda/\alpha \sim w_c$. **Disc source**: $\gamma = 2J_1(u\alpha/2)/(u\alpha/2)$, $w_c \approx 1.22\lambda/\alpha$. (Examples Q15 specialises this to a slit in a lens focal plane, $\gamma(d) = \mathrm{sinc}(\pi wd/f\lambda)$.)

10 Vector potential of standard configurations

Strategy: Use $\oint_L \mathbf{A}\cdot d\mathbf{L} = \Phi = \int \mathbf{B}\cdot d\mathbf{S}$ and the symmetry $d\mathbf{A} \parallel \mathbf{J}$.

Long straight wire. $\mathbf{A} = A_z\hat{z}$ with $A_z(r)$; using $B_\phi = \mu_0 I/2\pi r$ and integrating $\mathbf{A}$ round a thin rectangle of width dr, $-\partial_r A_z = B_\phi$, hence $A_z = -\dfrac{\mu_0 I}{2\pi}\ln r + \text{const}$.

Solenoid (flux method). $\oint A_\phi\,dl = A_\phi 2\pi r = \Phi$. Inside: $A_\phi = \mu_0 nIr/2$; outside: $A_\phi = \mu_0 nIa^2/(2r) \neq 0$ even though $\mathbf{B} = 0$ — the physical seed of the Aharonov–Bohm effect.

Distant loop (magnetic dipole). Expanding $|\mathbf{r} - \mathbf{r}'|^{-1}$ to first order in r'/r, the leading non-zero term uses $(\mathbf{r}' \times d\mathbf{r}') \times \mathbf{r}$ and a perfect-differential cancellation around the loop to give

$$\boxed{\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi}\frac{\mathbf{m}\times\mathbf{r}}{r^3}, \qquad \mathbf{m} = \tfrac{1}{2}\oint_L \mathbf{r}'\times d\mathbf{r}' = IA\hat{\mathbf{n}}.}$$

Examples Q16–Q17 invert this: given $\mathbf{B}(\mathbf{r})$ of a static dipole, reconstruct $\mathbf{A}$ (and check a candidate field is physical via $\nabla\cdot\mathbf{B} = 0$, then find $\mathbf{J} = \nabla\times\mathbf{B}/\mu_0$).

11 Gauge invariance in QM and the Aharonov–Bohm phase

Strategy: Show the Schrödinger equation keeps its form under a gauge change provided $\Psi \to e^{iq\chi/\hbar}\Psi$; then evaluate the path-dependent phase.

Starting from $\frac{1}{2m}[-i\hbar\nabla - q\mathbf{A}]^2\Psi + q\phi\Psi = i\hbar\dot{\Psi}$ and multiplying the trial $\Psi' = e^{iq\chi/\hbar}\Psi$ through, the identity $e^f\partial_y = (\partial_y - \partial_y f)e^f$ shows Ψ' obeys the same equation with $\mathbf{A}' = \mathbf{A} + \nabla\chi$, $\phi' = \phi - \dot{\chi}$. In a field-free region $\mathbf{A} = \nabla\chi$ with $\chi(\mathbf{r}) = \int_{\mathbf{r}_0}^{\mathbf{r}}\mathbf{A}\cdot d\mathbf{s}$, so $\Psi = \Psi'_{A=0}\exp\big(\frac{iq}{\hbar}\int\mathbf{A}\cdot d\mathbf{s}\big)$. For two paths enclosing flux Φ_B,

$$\boxed{\Delta = \frac{q}{\hbar}\oint\mathbf{A}\cdot d\mathbf{s} = \frac{q}{\hbar}\int_S(\nabla\times\mathbf{A})\cdot d\mathbf{S} = \frac{q\Phi_B}{\hbar}.}$$

With superconducting flux quantisation $\Phi = n\,h/2e$, $\Delta = n\pi$.

12 Maxwell's equations for the potentials; retarded solution

Strategy: Insert $\mathbf{E} = -\dot{\mathbf{A}} - \nabla\phi$, $\mathbf{B} = \nabla \times \mathbf{A}$ into ME1, ME4 and remove the cross terms with the Lorenz gauge.

ME1 gives $-\partial_t(\nabla\cdot\mathbf{A}) - \nabla^2\phi = \rho/\epsilon\epsilon_0$; ME4 gives $\nabla(\nabla\cdot\mathbf{A} + \frac{\epsilon\mu}{c^2}\dot{\phi}) - \nabla^2\mathbf{A} = \mu\mu_0\mathbf{J} - \frac{\epsilon\mu}{c^2}\ddot{\mathbf{A}}$. Imposing the Lorenz condition $\nabla\cdot\mathbf{A} + \frac{\epsilon\mu}{c^2}\dot{\phi} = 0$ decouples them into wave equations. A point time-varying charge has spherically symmetric outgoing solution $\phi \sim \frac{1}{r}g(t - r/c)$, matched to the static Poisson result for $r \ll c\tau$; superposition over a distribution gives

$$\boxed{\phi(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0}\int \frac{[\rho]}{|\mathbf{r}-\mathbf{r}'|}dV', \quad \mathbf{A}(\mathbf{r},t) = \frac{\mu_0}{4\pi}\int \frac{[\mathbf{J}]}{|\mathbf{r}-\mathbf{r}'|}dV', \quad [\,\cdot\,] \equiv t - \frac{|\mathbf{r}-\mathbf{r}'|}{c}.}$$

13 The Hertzian dipole fields (full derivation)

Strategy: Get $\mathbf{A}$ from the retarded current, then $\mathbf{B} = \nabla \times \mathbf{A}$ and $\mathbf{E} = -\dot{\mathbf{A}} - \nabla\phi$ (with the retarded ϕ); read off the $1/r^3, 1/r^2, 1/r$ ranges.

Step 1 (potential). For $d \ll r$, $|\mathbf{r}-\mathbf{r}'| \to r$ and $\int[\mathbf{J}]dV' = [\dot{\mathbf{p}}]$, so $\mathbf{A} = \dfrac{\mu_0}{4\pi r}[\dot{\mathbf{p}}]\,\hat{z}$, with polar components $A_r = A\cos\theta$, $A_\theta = -A\sin\theta$.

Step 2 (magnetic field). Only B_ϕ survives:

$$B_\phi = -\sin\theta\,\frac{\partial A}{\partial r} = \frac{\mu_0\sin\theta}{4\pi}\Big(\frac{[\dot{p}]}{r^2} + \frac{[\ddot{p}]}{rc}\Big),$$

using $\partial_r[F] = -\frac{1}{c}[F']$.

Step 3 (electric field). The retarded dipole potential (expanding $r_\pm \simeq r \mp \frac{d}{2}\cos\theta$) is $\phi = \dfrac{\cos\theta}{4\pi\epsilon_0}\big(\frac{[p]}{r^2} + \frac{[\dot{p}]}{rc}\big)$. Then $\mathbf{E} = -\dot{\mathbf{A}} - \nabla\phi$ yields

$$\boxed{\begin{aligned} E_r &= \frac{2\cos\theta}{4\pi\epsilon_0}\Big(\frac{[p]}{r^3} + \frac{[\dot{p}]}{r^2c}\Big),\\ E_\theta &= \frac{\sin\theta}{4\pi\epsilon_0}\Big(\frac{[p]}{r^3} + \frac{[\dot{p}]}{r^2c} + \frac{[\ddot{p}]}{rc^2}\Big), \qquad E_\phi = 0.\end{aligned}}$$

Step 4 (interpretation). The three terms scale as $1/r^3$ (near/static), $1/r^2$ (induction) and $1/r$ (radiation). In the far field only the radiation parts survive: $E_\theta = \dfrac{\sin\theta}{4\pi\epsilon_0}\dfrac{[\ddot{p}]}{rc^2}$, $B_\phi = E_\theta/c$ — transverse, in phase, linearly polarized along $\hat{\boldsymbol{\theta}}$.

13.1 Power radiated

The far-field Poynting flux $N = E_\theta B_\phi/\mu_0 \propto \sin^2\theta/r^2$. Integrating over a sphere and time-averaging $p = p_0e^{-i\omega t}$ ($\langle\ddot{p}^2\rangle = \frac{1}{2}\omega^4p_0^2$):

$$\boxed{\langle P\rangle = \frac{\mu_0\omega^4p_0^2}{12\pi c} = \frac{\omega^4p_0^2}{12\pi\epsilon_0c^3}, \qquad G_{ED}(\theta) = \tfrac{3}{2}\sin^2\theta.}$$

14 Multipole expansion: ED, MD and EQ terms

Strategy: Expand $e^{-ik\mathbf{n}\cdot\mathbf{r}'}$ in $\mathbf{A}(\mathbf{r})$; the m=0 term is the ED, and the m=1 term splits (anti)symmetrically into MD and EQ.

With $\mathbf{A}(\mathbf{r}) = \dfrac{\mu_0}{4\pi}\dfrac{e^{ikr}}{r}\sum_m \dfrac{(-ik)^m}{m!}\int \mathbf{J}(\mathbf{r}')(\mathbf{n}\cdot\mathbf{r}')^m dV'$:

- $m{=}0$: using $\nabla\cdot\mathbf{J} = i\omega\rho$ and integration by parts, $\int \mathbf{J}\,dV' = -i\omega\mathbf{p}$, giving electric-dipole radiation.
- $m{=}1$ antisymmetric: $\frac{1}{2}(\mathbf{r}'\times\mathbf{J})\times\mathbf{n}$ gives $\mathbf{A} = \dfrac{ik\mu_0}{4\pi}(\mathbf{n}\times\mathbf{m})\dfrac{e^{ikr}}{r}$ with $\mathbf{m} = \frac{1}{2}\int \mathbf{r}\times\mathbf{J}\,dV$ (magnetic dipole). Because $\mathbf{A}_{MD} \propto \mathbf{B}_{ED}$, the fields follow by the duality $c\mathbf{p}\to\mathbf{m}$.
- $m{=}1$ symmetric: electric quadrupole, with $Q_{\alpha\beta} = \int(3x_\alpha x_\beta - r^2\delta_{\alpha\beta})\rho\,dV$.

Comparative strengths (for current I_0, size a): $\langle P_{MD}\rangle/\langle P_{ED}\rangle \sim (2\pi a/\lambda)^2 \ll 1$, and $\langle P_{EQ}\rangle \propto \omega^6$.

15 Antenna reciprocity: $A_{\text{eff}} = \frac{\lambda^2}{4\pi}G$

Strategy: Place an antenna and a matched load in a black-body cavity at temperature T; equate absorbed and (Johnson-noise) re-radiated power (detailed balance).

Using the Rayleigh–Jeans density $U(\nu) = 8\pi\nu^2 k_B T/c^3$ (one polarization, so halve it), the power absorbed from solid angle $d\Omega$ is $P_{\text{abs}} = A_{\text{eff}}\frac{\nu^2}{c^2}k_B T\,d\Omega\,d\nu$. The load's Nyquist noise $\langle V^2\rangle = 4k_B T R_r\,d\nu$ drives $\langle I^2\rangle = k_B T\,d\nu/R_r$, radiating $P_{\text{rad}} = \langle I^2\rangle R_r\frac{G}{4\pi}d\Omega = k_B T\frac{G}{4\pi}d\Omega\,d\nu$. Equating $P_{\text{abs}} = P_{\text{rad}}$:

$$\boxed{A_{\text{eff}}(\theta,\phi) = \frac{\lambda^2}{4\pi}G(\theta,\phi) \qquad \text{(for any antenna).}}$$

The radiation resistance of a Hertzian dipole follows from $R_r = \langle P\rangle/\langle I^2\rangle$ with $\dot{p} = Id$: $R_r = \frac{2\pi}{3}Z_0(d/\lambda)^2 \approx 789(d/\lambda)^2\,\Omega$.

16 Scattering: Thomson, Rayleigh, and the structure factor

Strategy: Drive the scatterer's dipole with the incident field; the radiated power over the incident flux is the cross-section.

Thomson (free electron). $m_e\ddot{z} = -eE_0e^{-i\omega t}$ gives $\ddot{p} = (e^2/m_e)E_0e^{-i\omega t}$; with $\langle P\rangle = \mu_0\langle\ddot{p}^2\rangle/6\pi c$ and incident flux $\frac{1}{2}c\epsilon_0E_0^2$,

$$\boxed{\sigma_T = \frac{\mu_0^2e^4}{6\pi m_e^2} = \frac{8\pi}{3}r_e^2 = 6.65\times10^{-29}\,\text{m}^2 \quad \text{(independent of } \omega\text{).}}$$

Rayleigh (induced $p_0 = \alpha E_0$). $\sigma = \mu_0^2\omega^4\alpha^2/6\pi \propto \lambda^{-4}$. **Scattered polarization.** For unpolarized incidence the degree of polarization at angle α is $P = (1-\cos^2\alpha)/(1+\cos^2\alpha)$. **Many scatterers.** $P_{\text{tot}} = P_1(\mathbf{q})\,|\sum_j e^{i\mathbf{q}\cdot\mathbf{r}_j}|^2$, $\mathbf{q} = \mathbf{k}_i - \mathbf{k}_s$; the structure factor $F(\mathbf{q}) = |\tilde{n}(\mathbf{q})|^2$. Random positions (random walk) give $\langle|\sum e^{i\mathbf{q}\cdot\mathbf{r}_j}|^2\rangle = N$, so $P_{\text{tot}} = NP_1$; correlated positions give Bragg peaks at reciprocal lattice vectors ($\mathbf{k}_s = \mathbf{k}_i - \mathbf{G}$).

17 Relativistic electrodynamics

17.1 The 4-current is a 4-vector

Strategy: Charge invariance + length contraction.

In the rest frame $\rho_0 = Q/V_0$; in S the volume contracts $V = V_0/\gamma$ so $\rho = \gamma\rho_0$. Then $J = (c\rho,\mathbf{J}) = \rho_0\frac{d}{d\tau}(ct,\mathbf{r})$, a 4-vector since ρ_0 and τ are invariant and $(ct,\mathbf{r})$ is a 4-vector. Contracting with ∂_α recovers $\partial_\alpha J^\alpha = 0$ in every frame.

17.2 Covariant Maxwell equations

With $A = (\phi/c, \mathbf{A})$ and $\Box^2 A = \mu_0 J$, defining $F^{\alpha\beta} = \partial^\alpha A^\beta - \partial^\beta A^\alpha$:

$$\partial_\alpha F^{\alpha\beta} = \mu_0 J^\beta, \qquad \partial_\alpha F_{\beta\gamma} + \partial_\beta F_{\gamma\alpha} + \partial_\gamma F_{\alpha\beta} = 0.$$

The field transformations (from $F'^{\alpha\beta} = \Lambda^\alpha_\gamma \Lambda^\beta_\delta F^{\gamma\delta}$) are $E'_\parallel = E_\parallel$, $\mathbf{E}'_\perp = \gamma[\mathbf{E} + \mathbf{v} \times \mathbf{B}]_\perp$, $B'_\parallel = B_\parallel$, $\mathbf{B}'_\perp = \gamma[\mathbf{B} - \frac{1}{c^2}\mathbf{v} \times \mathbf{E}]_\perp$. *Examples Q30* shows $|\mathbf{E}|^2 - c^2|\mathbf{B}|^2$ and $\mathbf{E} \cdot \mathbf{B}$ are invariant.

17.3 Magnetism as a relativistic effect

Strategy: Compute the electrostatic force on a charge moving parallel to a neutral wire in the carriers' frame, then transform the force back.

The moving negative carriers, un-contracted in their rest frame, have density $\rho'_- = -ne/\gamma_v$. Transforming the resulting electrostatic force back to S with $f_y = f'_y \gamma_v (1 - vu/c^2)$ and adding the (Gauss-law) force of the positive background gives a net $f_y = \dfrac{\mu_0(-I)}{2\pi r} qu$ — exactly the Lorentz force from the wire's $\mathbf{B}$. The same field in the electron's rest frame ($\mathbf{B}' \sim -\frac{1}{c^2}\mathbf{v} \times \mathbf{E}$) produces **spin–orbit coupling** (with the $\frac{1}{2}$ Thomas factor).

18 Radiation from moving charges

18.1 Uniformly moving charge does not radiate

Transforming the rest-frame Coulomb field gives $E_\perp/E_\parallel = \tan\theta$ (field points from the present position) and $B_\phi = (v/c^2)E_\perp$. The flux $\mathbf{N} \sim 1/r^4$, so no energy escapes: the field energy is merely carried along. For $\gamma \gg 1$ the field is compressed into the transverse plane ($E_\perp \sim \gamma\, q/4\pi\epsilon_0 r^2$).

18.2 Čerenkov angle

Strategy: Huygens construction with wavelets of speed c/n from earlier positions.

For $v > c/n$ the wavelets' common tangent makes an angle

$$\cos\theta_C = \frac{cT/n}{vT} = \frac{c}{nv} = \frac{1}{\beta n}.$$

The Fourier (Frank–Tamm) treatment of MA1/MA2 with sources $\rho = q\delta(\mathbf{r} - \mathbf{v}t)$ shows energy escapes to infinity only where $\kappa^2 = \frac{\omega^2}{v^2}[1 - \beta^2\epsilon(\omega)]$ is imaginary, i.e. $\epsilon(\omega) > 1/\beta^2$ — the same criterion — giving emission in bands (the characteristic blue glow).

18.3 Liénard–Wiechert potentials and fields

Strategy: Integrate the covariant retarded Green's function over the particle world line using $\delta[(x - r(\tau))^2]$.

Performing the τ-integral with the light-cone condition gives

$$\Phi = \frac{1}{4\pi\epsilon_0}\left[\frac{e}{(1 - \boldsymbol{\beta} \cdot \mathbf{n})R}\right]_{\text{ret}}, \quad \mathbf{A} = \frac{\mu_0}{4\pi}\left[\frac{e\mathbf{v}}{(1 - \boldsymbol{\beta} \cdot \mathbf{n})R}\right]_{\text{ret}}.$$

The fields split into velocity ($\propto R^{-2}$) and acceleration ($\propto R^{-1}$) parts:

$$\mathbf{E} = \frac{e}{4\pi\epsilon_0}\left\{\frac{\mathbf{n}-\boldsymbol{\beta}}{\gamma^2(1-\boldsymbol{\beta}\cdot\mathbf{n})^3R^2} + \frac{\mathbf{n}\times[(\mathbf{n}-\boldsymbol{\beta})\times\dot{\boldsymbol{\beta}}]}{c(1-\boldsymbol{\beta}\cdot\mathbf{n})^3R}\right\}_{\text{ret}}, \qquad \mathbf{B} = \tfrac{1}{c}[\mathbf{n}\times\mathbf{E}]_{\text{ret}}.$$

18.4 Larmor and Liénard power

Strategy: For $v \ll c$ keep only the acceleration field; for general v use that the radiated power is a Lorentz invariant and the acceleration 4-vector.

Non-relativistic Larmor: $P = \dfrac{\mu_0 e^2}{6\pi c}|\dot{\mathbf{v}}|^2$. Since P is invariant and $\alpha^2 \equiv -\text{Acc}\cdot\text{Acc} = \gamma^6 a_\parallel^2 + \gamma^4 a_\perp^2$,

$$P = \frac{\mu_0 e^2}{6\pi c}\big(\gamma^6 a_\parallel^2 + \gamma^4 a_\perp^2\big) \qquad \text{(Liénard).}$$

18.5 Synchrotron beaming and pulse width

Strategy: In circular motion $a = a_\perp$; the lab-frame $(1-\boldsymbol{\beta}\cdot\mathbf{n})$ factors beam the radiation into a $1/\gamma$ cone, so the observer sees a short pulse.

The angular power in the charge's own time carries a $(1-\boldsymbol{\beta}\cdot\mathbf{n})^{-5}$ factor, confining radiation to $\theta \lesssim 1/\gamma$. The pulse seen from an arc of length $2R/\gamma$ has duration

$$\Delta t \sim \frac{2R}{\gamma}\Big(\frac{1}{v} - \frac{1}{c}\Big) = \frac{2}{\gamma\omega_B}\Big(1 - \frac{v}{c}\Big) \sim \frac{1}{\gamma^3\omega_B},$$

so the spectrum is broad up to $\nu_s \sim \gamma^3\omega_B$ with $\lambda_s \sim R/\gamma^3 \ll R$ (hard X-rays). The gyrofrequency is $\omega_B = eB/\gamma m$ and the instantaneous power $P = \mu_0 e^4\gamma^2 B^2 v^2/6\pi c m^2$. In the cyclotron limit $(\gamma \to 1)$ the motion is two quadrature dipoles radiating *monochromatically* at ω_B.

19 Selected examples-sheet methods worth memorising

- **Cavity modes (Q11).** Write the field as $\pm z$-going waves in the polarization eigenstates of the medium; boundary conditions at $z = 0, L$ quantise each $k = n_\star\omega/c$. Birefringent: $\omega = \pi c\ell/(n_o L)$ and $\pi c\ell/(n_e L)$. Optically active / Faraday: split modes $\omega = \frac{\pi c}{nL}\ell \pm$ (rotatory / VBc/n) terms.
- **Helicon modes (Q12).** Low-frequency limit of $\epsilon_{L,R}$ gives $\omega = c^2k^2\omega_c/\omega_p^2$ and $v_g = 2c\sqrt{\omega\omega_c}/\omega_p$; only the LCP mode propagates.
- **Pulsar spin-down (Q20).** A rotating magnetic dipole radiates $\propto \ddot{m}^2 \propto \omega^4 M^2$; energy balance with rotational kinetic energy gives $\omega\ddot{\omega} = 3\dot{\omega}^2$. Crab data then fix M and the surface field ($\sim 10^9$ T).
- **Loop antenna (Q21).** From the quoted loop power $\mu_0 I_0^2\omega^4 a^4/(12\pi c^3)$: $R_r = \mu_0\omega^4 a^4/(6\pi c^3)$, $G = \frac{3}{2}\sin^2\theta$, and $A_{\text{eff}} = \frac{\lambda^2}{4\pi}G$ gives the combined scattering+absorption cross-section $3\pi c^2/\omega^2$.
- **Moving dielectric slab (Q24).** Relativistic velocity addition of the in-medium light speed gives $n_v = (n+\beta)/(n\beta+1)$ and a lateral ray displacement — a relativistic aberration calculation.
- **Co-moving charges (Q29).** Use the uniformly-moving-charge fields: forces F/γ^2 (along motion), F/γ (transverse), with $F = q^2/4\pi\epsilon_0 r^2$.

End of derivations. See the companion "Course Summary" for the conceptual overview and where each result fits.